MIL - STD- 1871(AT)

20 April 1983

Superseding
(see 6.1)

MILITARY STANDARD

ROTOR, TURBINE, CAST MATERIAL
ACCEPTANCE CRITERIA FOR

FSC 2835

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DEPARTMENT OF DEFENSE
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Rotor, Turbine, Cast, Material Acceptance Criteria for

1. This Military Standard is approved for use by US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: DRSTA-GSS, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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FOREWORD

Procedures covered by this standard are used to provide acceptance criteria for material in a specific shape (turbine rotor) for use in a specific application. In addition, this standard defines rework limitations.

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1. SCOPE

1.1 Scope. This standard covers material acceptance criteria for turbine rotor castings and finish machined turbine rotors.

1.2 Classification. 1/ Casting classification designated on the engineering drawing for purposes of inspection shall be in accordance with the following:

MIL-STD-1871-P1	Castings not subjected to the hot isostatic pressing (HIP) process
MIL-STD-1871-P2	Castings subjected to the HIP process

1/. When no classification (suffix) is specified, castings shall be inspected in accordance with MIL-STD-1871-P1.

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2. REFERENCED DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this standard to the extent specified herein.

SPECIFICATIONS
MILITARY

MIPD-H-62411	- Hot isostatic pressing, procedure for.
MIPD-STD-1872	- Radiographic inspection.
MIPD-P-62434	- Penetrant inspection.

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from U.S. Army Tank-Automotive Command, ATTN: DRSTA-GSS, Warren, MI 48090, or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this standard to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E192	- Investment Steel Castings for Aerospace Applications.
ASTM E513	- Constant-Amplitude, Low Cycle Fatigue Testing.

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, PA 19103.)

AVCO LYCOMING DIVISION

TE 24322	- Low Cycle Fatigue Specimen.

(Application for copies should be addressed to Avco Lycoming Division, Stratford, CT 06497.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. DEFINITIONS

3.1 Definitions. For purposes of this standard, the following definitions shall apply.

3.1.1 Low cycle fatigue (LCF) terminology. LCF terms are defined in Standard ASTM E513.

3.1.2 HIP lot. A HIP lot shall consist of all those castings from one casting lot which have been hot isostatic pressed in one HIP container at one time.

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4. GENERAL REQUIREMENTS

4.1 Acceptance criteria.

4.1.1 Imperfections (visual and flourescent penetrant). Limits for imperfections shall be as follows:

a. There shall be no evidence of cracks, dross, cold shuts, hot tears, and linear or crack-like indications.

b. Mold ridges, fins, and gate witnesses shall not exceed 0.030 inch in height in the disc and hub area. There shall be no mold ridges, or positive metal in excess of 0.005 inch in height in the hot gas airflow area.

c. Pattern assembly seal lines shall not exceed + 0.015 inch from nominal feature contours.

d. Unrateable indications, pits, blowholes, bottomed porosity, clusters of indications, etc, shall be within the allowable limits specified in table I (see figures 1, 2 and 3).

e. The limits of table I may be exceeded on as-cast surfaces provided the allowable limits of table I can be met after machining.

f. Coating shall be free of chips, dents, burnish marks or any degradation below the allowable coating thickness as follows:

(1) Blades

(a) Leading edge: 2.0 to 4.0 mils.
(b) Trailing edge: 1.5 to 3.5 mils.
(c) Airfoil surface: Coating thickness shall be a gradual change from leading to trailing edge without any sharp changes either depressed or elevated.

(2) Other surfaces: 1.5 to 4.0 mils.

4.1.2 Indications (radiographic). Indications shall be within the allowable limits specified in table II (see figures 1, 2 and 3).

4.1.3 Grain size. Grain size shall be in accordance with engineering drawing requirements.

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TABLE I. Visual and flourescent penetrant acceptance limits

Type of indication	Casting area A	Casting area B	Casting area C
Unrateable	0.010 inch major dim.	0.015 inch major dim.	0.020 inch major dim.
Pits, blowholes, bottomed porosity	0.015 inch major dim. Dist of closest approach 5X major dim of larger indication. Max number - no limit.	0.025 inch major dim. Dist of closest approach 5X major dim of larger indication. Max number - no limit.	0.050 inch major dim. Dist of closest approach 0.125 inch. Max number - no limit.
	Over 0.015 to 0.030 inch major dim. Dist of closest approach 0.250 inch. Max number - 2 per blade	Over 0.025 to 0.040 inch major dim. Dist of closest approach 0.250 inch. Max number - 2 per blade	
Clusters of indications			
Individual indication in cluster	0.020 inch major dim.	0.020 inch major dim.	0.030 inch major dim.
Max size of cluster	0.100 inch major dim.	0.100 inch major dim.	0.15 inch major dim.
Dist between clusters	0.50 inch min.	0.50 inch min.	0.30 inch min.

1. Clusters or single indications shall not be opposite or relateable.
2. Rateable indications shall not break over an edge.
3. Where two surfaces of different classifications adjoin at a corner, allowable indications on the casting area B or C surface must be 0.010 inch from the edge which adjoins a casting area A surface.
4. A cluster is a group of rateable and/or unrateable indications within the specified area. The number of indications present shall not exceed eight.

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TABLE II. Radiographic acceptance limits

Indications	ASTM E192 reference number	Casting area A	Casting area B	Casting area C
Hot tears	4.1		Rejectable	
Cracks	4.2		Rejectable	
Cold shuts	4.3		Rejectable	
Gas holes				
1/8"	1.1	2	3	(1)
3/4"	1.1	(1)	(2)	3
Shrinkage cavity	2.1	(2)	(2)	2
Shrinkage sponge	2.2			
1/8"		2	2	(1)
3/4"		(1)	1	1
Shrinkage dendritic	2.4			
1/8"		1	2	(1)
3/4"		(1)	1	3
Foreign material Less dense	3.11			
1/8"		2	3	(1)
3/4"		(1)	3	2
Foreign material More dense	--	(3)	(3)	(3)

Notes:

(1) Not applicable.

(2) None acceptable.

(3) Use plate 3.11 (foreign material less dense) for size and frequency only. Do not consider apparent density.

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4.1.4 Physical and mechanical properties. Physical and mechanical properties of castings shall meet all requirements of the material specification with the following exemptions for HIP processed castings.

4.1.4.1 Surface condition. When a specimen is examined at a magnification of 500X, evidence of any surface condition resulting from a reaction of the metal with the HIP environment shall not exceed the following:

a. There shall be no evidence of penetration of any surface reactions greater than that permitted in the material specification for surface oxides.

b. There shall be no evidence of any intergranular attack.

4.1.4.2 Microstructure. Microstructure specimens shall be examined at a magnification of 100X for microporosity and/or microshrinkage in a manner approved by the acquisitioning activity. When examined at specified magnification, the limits of acceptability shall be as follows:

a. Maximum dimension for individual microporosity or microshrinkage within a field - 0.003 inch (0.076 mm).

b. Unrateable - 0.001 inch (0.025 mm) maximum dimension.

c. Two or more imperfections shall be rated as a single indication if the distance between indications is less than the maximum dimension of larger indication.

d. Total cumulative length of all rateable indications in a single field shall not exceed 0.010 inch (0.254 mm).

e. Field diameter shall be 0.040 inch (1.02 mm).

4.1.4.3 Low cycle fatigue life. LCF specimen referenced on Drawing TE 24322, shall withstand a low cycle fatigue test of 1000 cycles minimum under the following conditions:

a. Strain range: 0.006 inch per inch (0.006 mm/mm).

b. Strain ratio: A = 1.0.

c. Function: Sinusoidal (Haversine).

d. Frequency: 0.2 Hertz (Hz).

e. Temperature: $900^{\circ} \pm 5^{\circ}F$ ($482^{\circ} \pm 2.8^{\circ}C$).

4.1.4.4 Tensile elongation. Tensile elongation of casting material shall average 5 percent minimum with no individual specimen being lower than 4 percent.

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5. DETAIL REQUIREMENTS

5.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the vendor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the vendor may utilize his own facilities or any other facility acceptable to the acquisitioning activity. The acquisitioning activity reserves the right to perform or witness any of the necessary tests to assure that turbine rotor material conforms to the requirements of this standard.

5.2 Test specimens. Unless otherwise specified, test specimens shall be in accordance with requirements of the material specification.

5.3 Reports. 2/ Unless otherwise specified on purchase order, supplier shall furnish three (3) copies of a certificate with each shipment stating following information:

a. Purchase order number.

b. Part number and revision letter.

c. Quantity of parts.

d. Serial numbers.

e. HIP lot number(s) (see 3.1.2).

f. Statement of conformance to MIL-STD-1871 latest revision.

2/ The receiving report shall include casting lot number and HIP lot number with a correlation to casting serial number.

5.4 Test methods.

5.4.1 Visual and flourescent penetrant. To verify conformance to 4.1.1, castings and finish machined rotors shall be visually examined and fluorescent penetrant inspected in accordance with M1PD-P-62434, Method P2.

5.4.2 Radiographic. To verify conformance to 4.1.2, castings shall be radiographically inspected in accordance with M1PD-STD-1872.

5.4.3 Grain size. To verify conformance to 4.1.3 (grain size determination), castings shall be examined after etching as follows:

a. Solution (at room temperature).

(1) Hydrogen peroxide (H_2O_2 reagent grade, 35 percent) - ten parts by volume.

(2) Hydrochloric acid (HCl reagent grade, 37 percent) - forty-five parts by volume.

(3) Water - forty-five parts by volume.

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b. Casting shall be immersed in etchant solution for 12 to 20 seconds.

c. Casting shall be thoroughly rinsed in clean cold running water.

d. Casting shall be dried with clean dry filtered air blast.

5.4.4 Destructive analysis. To verify conformance to physical and mechanical properties requirements of 4.1.4, destructive analysis shall be performed as specified in material specification.

5.4.4.1 Microporosity analysis. To verify conformance to 4.1.4.2, every HIP run shall contain a minimum of one test specimen to determine effectiveness of HIP process through depletion of specimen microporosity. Size, shape and condition of specimen shall be as approved by acquisitioning activity.

5.4.4.2 Low cycle fatigue test. To verify conformance to 4.1.4.3, the low cycle fatigue specimen shall be subjected to a low cycle fatigue test under conditions specified (see 3.1.1).

5.4.4.3 Specimen failure. HIP cycled castings with evidence of subsurface microporosity or microshrinkage, or with mechanical property failures shall be dispositioned as follows:

a. The presence of microporosity or microshrinkage (ref 4.1.4.2) in a test specimen or macroslice shall subject HIP lot represented by specimen to metallurgical review. Castings represented by failed test specimens may be reworked in accordance with 5.5 provided castings are found to be otherwise metallurgically sound.

b. When a test specimen fails mechanical property requirements, but shows no evidence of unacceptable microporosity or microshrinkage (ref 4.1.4.2), lot of castings represented shall be retested in accordance with requirements of the material specification.

5.5 Rework. Castings or finish machined turbine rotors which exceed, the limits of 4.1.1 and 4.1.2 may be reworked to extent specified herein.

5.5.1 Polishing. Removal of positive metal not specifically permitted in 4.1.1.b and 4.1.1.c shall be accomplished by polishing.

5.5.2 Blending. 3/ Imperfections and indications (negatives) shall be smoothly blended into surrounding area by means of a carbide burring bit or equivalent tool. All blending shall be within limits specified in table III (see figure 1).

3/ For coated surfaces requiring blending, entire part shall be stripped of coating prior to blending operation.

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TABLE III. Blend limits

Area	Blend limits
Casting Area A	
Blade leading edge	0.025 inch max depth, one per blade. 0.060 inch max depth, one per blade; 10 blades per wheel.
Blade trailing edge	Same as leading edge.
Leading edge plus trailing edge	Chordal reduction shall not exceed 20 percent if blends on same blade are radially aligned.
Lower airfoil surface	0.125 inch max dimension; depth - 0.020 or 20 percent of section which ever is less; one per blade; 10 blades per wheel.
Platform and shroud fillets	0.030 inch max depth. One per each blade and no wheel limit; no notching or reduction of fillet radius.
Casting Area B	
Airfoil surface	0.187 inch max dim; depth - 0.030 inch or 30 percent of section whichever is less; one per blade; no limit per wheel.
Shroud surfaces	0.187 inch max dim; depth - 0.030 inch or 30 percent of section whichever is less; no limit per wheel.
Hub area (machined)	Any blending which will not exceed engineering drawing tolerances.
Platform surface	0.250 maximum dimension, 0.030 inch max depth; 10 places per wheel.
Casting Area C	
Disc area (machined)	Any blending which will not exceed engineering drawing machining tolerances.

Notes:

1. Unless otherwise specified, blend widths shall be 8 to 20 times blend depth.
2. All blends shall be smoothly blended into surrounding surfaces.
3. Blends that overlap into an adjacent area shall be evaluated to the limits of the most critical area.

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5.5.3 Reprocessing. Castings (HIP) may be reprocessed in accordance with requirements of MIPD-H-62411.

6. NOTES

6.1 Supersession. This military standard supersedes AVCO Lycoming Inc. internal specification P7123E, dated 30 April 1979.

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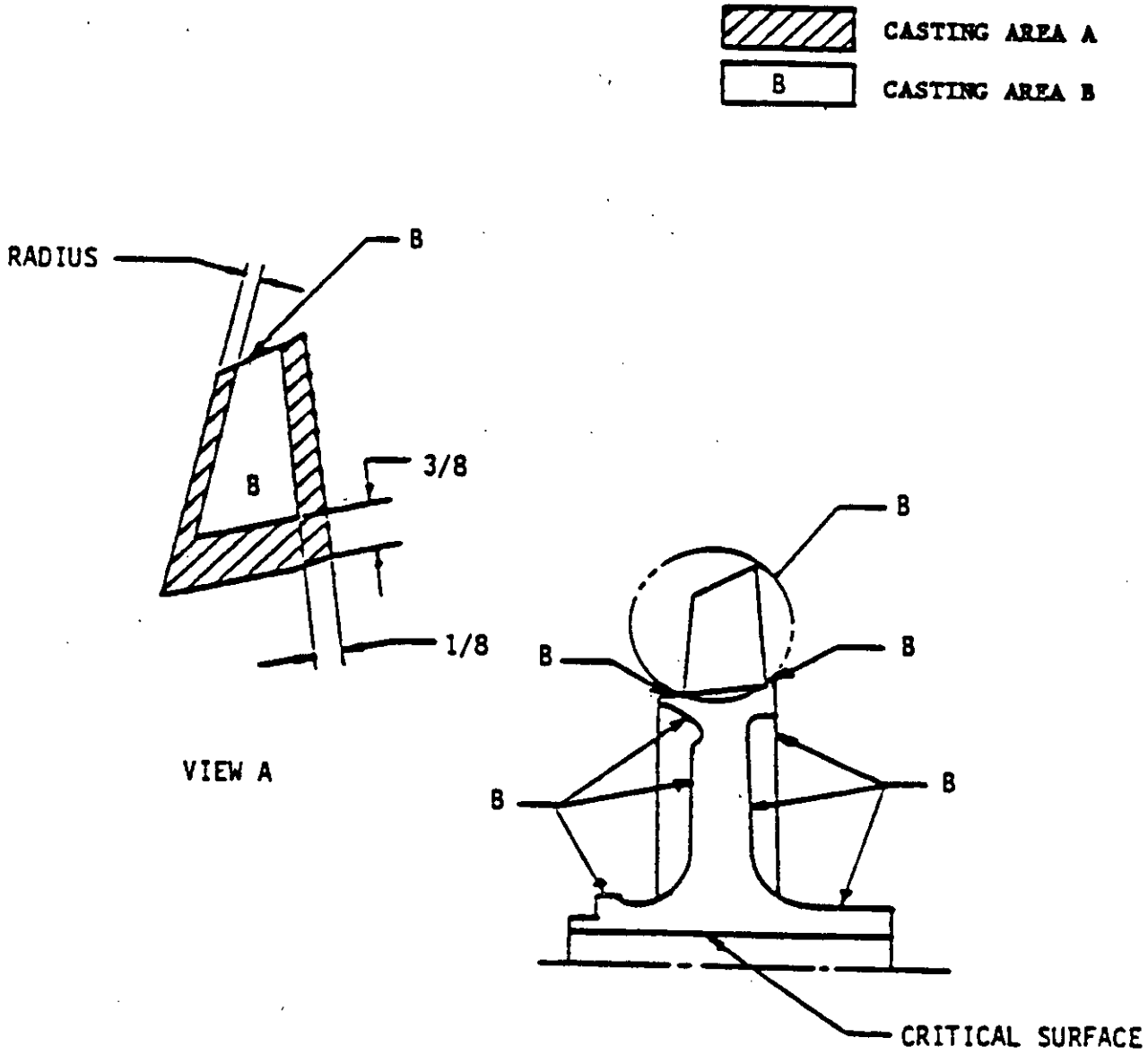


FIGURE 1. Casting areas on rotor

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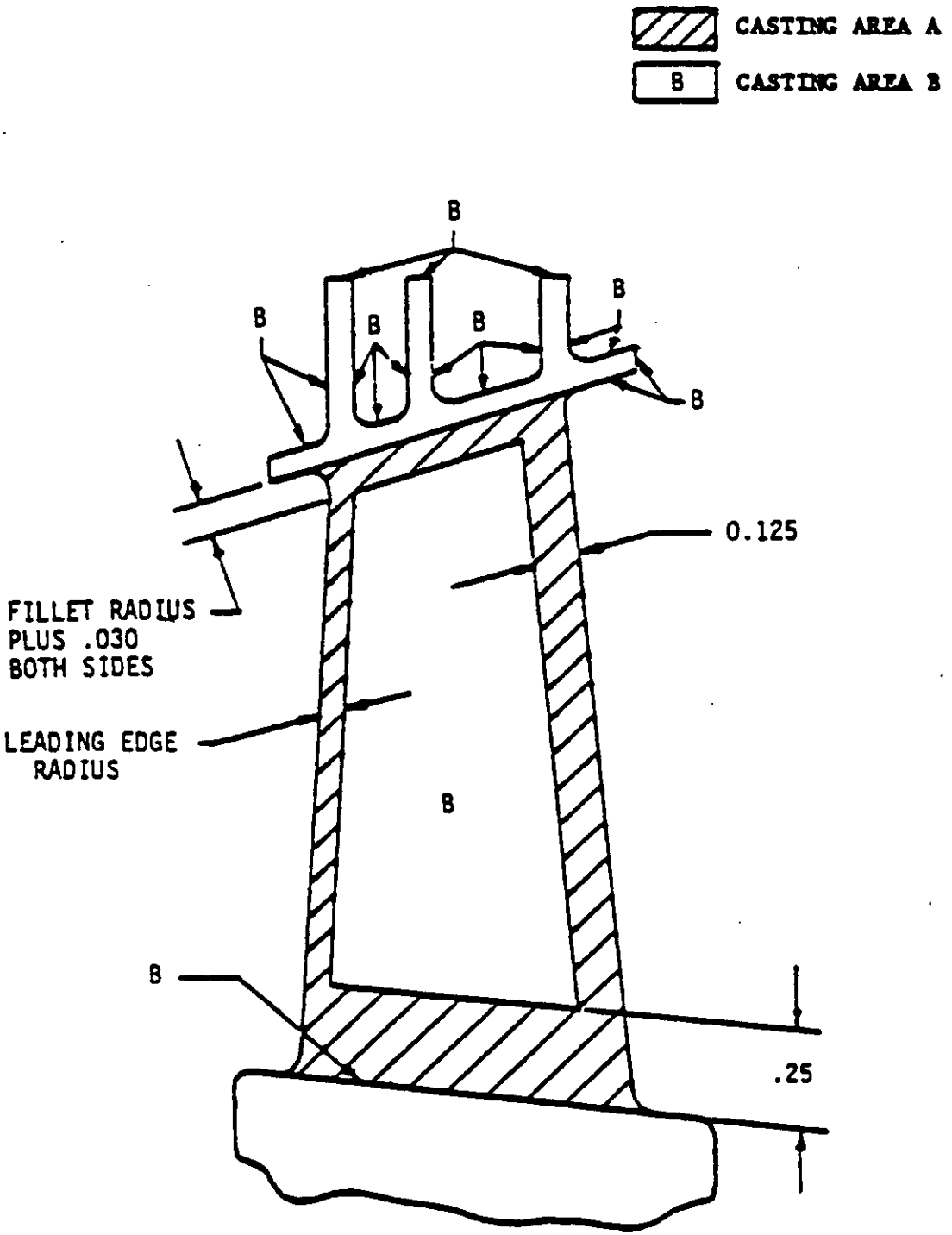


FIGURE 2. Casting areas of rotor blades

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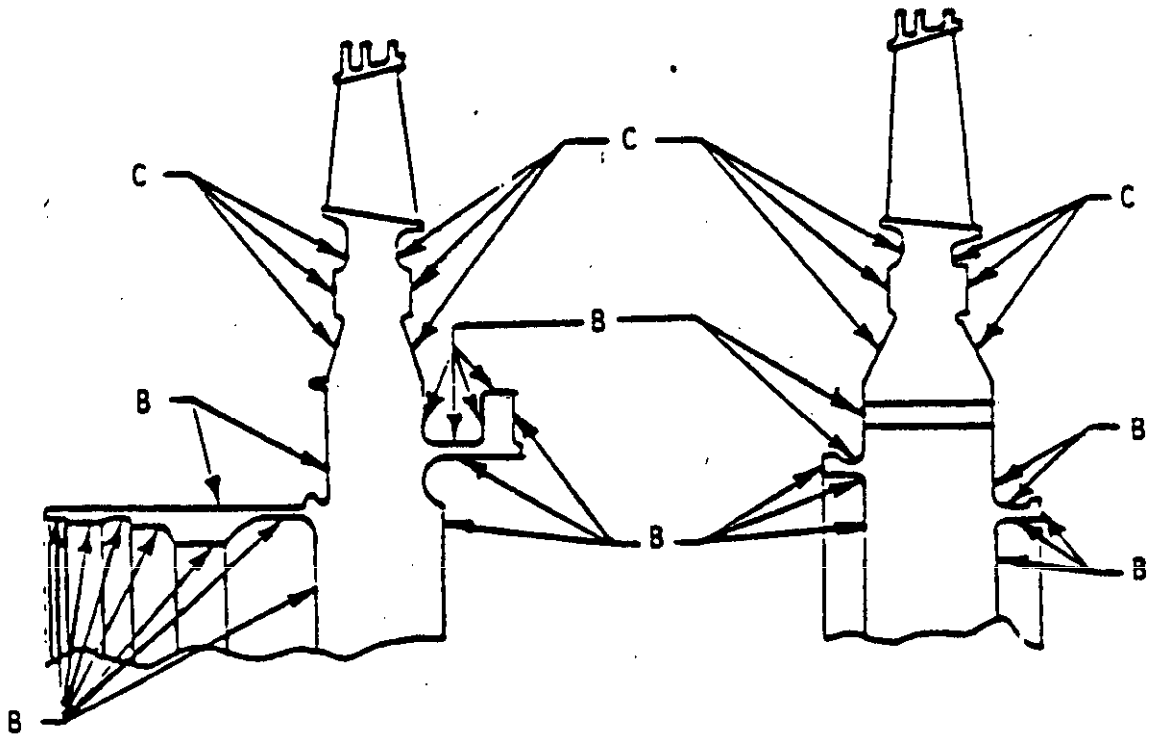


FIGURE 3. Casting areas of rotor disc

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